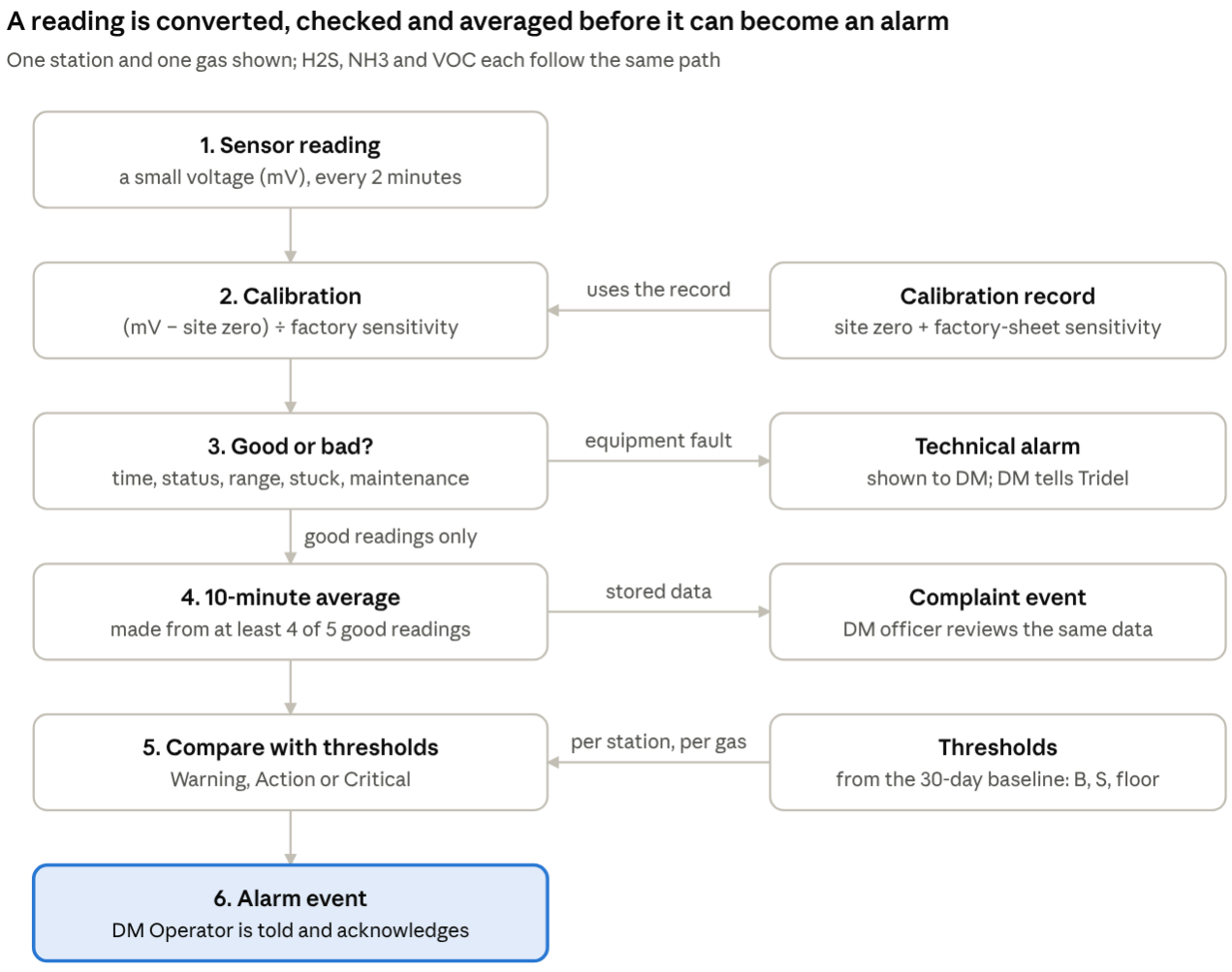
AQMS Handover Note for the Terra team

Nov 4, 2026 · @Amandeep

This note explains, in plain words, how the five AQMS stations at the Al Qusais landfill work, so the Terra team can build the set-up screens, calibration, alarms and events without needing to call me. It uses only what Dubai Municipality asked for, what Alphasense (the sensor maker) gives (its formulas, datasheets and its own advice to us), and, until the stations measure their sensor temperature, the Open-Meteo weather service. I am away from Nov 6, 2026 to Nov 16, 2026 .

The whole system in one picture



How one reading becomes an alarm · 6 steps, 4 side records

Read it top to bottom. A reading is converted, checked, averaged and compared, and only then can it become an alarm. Equipment faults become technical alarms that DM sees in Terra, and DM tells Tridel, and a smell complaint looks at the same stored data without being started by an alarm.

Words you will see

Read this table once. Every other section uses these words and nothing harder.

Word	What it means, simply
Station	One box on a pole at the landfill. There are five, named AQMS-01 to AQMS-05.
Sensor	The small part inside a station that "smells" one gas.
H2S	Hydrogen sulphide, the rotten-egg smell. Shown in **ppb**.
NH3	Ammonia, the sharp toilet-cleaner smell. Shown in **ppm**.
VOC	Many smelly chemicals measured together by one sensor (the PID). Shown in **ppm, as isobutylene** — isobutylene is just the reference gas it is compared against.
ppm and ppb	Parts per million and parts per billion. 1 ppm = 1,000 ppb.
mV	Millivolt. The tiny electric signal a sensor gives out.
WE (working electrode)	The part of the sensor that touches the gas. Its signal goes up when gas is there.
AE (auxiliary electrode)	A twin part that never touches the gas. It only feels heat, so we use it to remove the heat effect. H2S has one; NH3 does not.
Offset or zero	What the sensor reads when there is **no** gas. We always subtract it.
Sensitivity	How many mV the sensor gives for each ppb or ppm of gas. Printed on the sensor's factory calibration sheet.
Certificate	The paper that comes with each sensor, with that sensor's own numbers on it.
Reading (record)	One value, taken every **2 minutes**.
10-minute average	One value made from the five 2-minute readings in those 10 minutes. Alarms use this.

Word	What it means, simply
Valid / invalid	Good reading / bad reading. Bad readings are kept but never used in averages or alarms.
Baseline	The normal level of a gas at one station, learned in the first 30 days.
Threshold	The line that, when crossed, starts an alarm.
Alarm	The system saying "this gas is higher than normal here".
Event	A record that is created when an alarm starts, or when Dubai Municipality (DM) registers a smell complaint.
Acknowledge	A person clicks "I have seen this". It does **not** stop the alarm.
UTC	World time. We store all times in UTC and show UAE time (UTC + 4 hours) on screen.
Site zero	The sensor's own reading in clean air, measured at the station. We use it as the zero instead of the factory zero, because real air, heat and our own board shift it.
Sensor fault	The sensor is not giving a real signal (for example, a voltage stuck at the end of its range). No number is shown.

Set-up (configuration)

Set-up means typing in, once, the facts about each station and each sensor. Only the **DM Administrator** can change set-up, because set-up changes what a number means.

1. For each station

Setting	What to enter	Example
Station ID	AQMS-01 to AQMS-05. This is how incoming data is matched to a station.	AQMS-03

Setting	What to enter	Example
Name and location	A name, latitude and longitude.	From the site survey
Channels	Which things this station measures.	H2S, NH3, VOC, wind speed, wind direction, temperature, humidity
Reading interval	How often a reading is taken.	2 minutes
Offline time	How long with no data before the "communication lost" alarm.	30 minutes
Data completeness	How much good data a report period needs.	75 %
Averages to keep	Which longer averages to make.	10 min, 1 hour, 8 hours, 24 hours
Alarm recipients	Who is told about alarms, per level.	DM Operator (in the app), plus an email list

2. For each sensor

Each sensor gets a calibration record: its sensitivity from the factory calibration sheet, and its zero (from a site zero, or from the factory sheet). The next section explains every number.

3. The allowed range of each channel

Terra uses these to mark a reading as "out of range". A reading outside its range is kept, but marked invalid.

Channel	Sensor type	Allowed range	Unit
H2S	Electrochemical, 4 electrodes	0 to 100 ppm	ppb on screen
NH3	Electrochemical, 3 electrodes	0 to 100 ppm	ppm
VOC	PID (light-based)	0 to 40 ppm	ppm as isobutylene
Wind speed	Ultrasonic	0 to 60 m/s	m/s
Wind direction	Ultrasonic	0 to 360	degrees

4. What a station sends

A station does **no** maths. It measures every 2 minutes and keeps the readings in its own memory. Every **10 minutes** it sends one package to Terra with the raw readings. If the network is down, it stores the readings and sends them all when the network comes back.

Each 10-minute package carries, for each of its five 2-minute readings:

- **H2S:** the WE voltage and the AE voltage, in mV;
- **NH3:** the WE voltage, in mV;
- **VOC:** the PID voltage, in mV;
- the temperature at the top of each gas sensor, if fitted (used for the temperature correction), and air temperature and humidity, if fitted, for fault-finding only;
- wind speed and wind direction;
- station ID, measurement time, set-up version, sensor and logger status (including warm-up, calibration or maintenance mode), power status and communication status.

Terra does everything else: turning voltages into gas, deciding good or bad, the 10-minute average, minimum and maximum, the completeness count, and the alarms. If a station sends volts instead of millivolts, multiply by 1,000 first. The Monitoring Criteria document lists averages inside the station package; because Terra now does the maths, Terra makes them instead.

When a package arrives, Terra checks the checksum, the number of fields, and that every value is a number. Keep the raw package text exactly as it arrived; it is needed for fault-finding.

4a. The record the test station sends today

The test station does not yet send the 10-minute packages described above. Today it sends **one record about every minute**, as a small block of text in JSON format (the station's "schema 4"). Terra must be able to read this record. Here is a real one, from 3 November 2026 at 09:59:10 UTC (13:59:10 in Dubai). The same record is in the package as `Station_record_example_schema4.json`, and the Excel sheet "Station record" works through it.


```
{
  "a0Volts": 0.36734,
  "adcOk": true,
  "averageCount": 1,
  "batteryV": 11.7587,
  "combinedString": "2026-11-03,09:59:10,0.36734,2.55422,2.55391,0.09812,NA",
  "destination": "client",
  "device": "aqms-monitor",
```

```
  "h2sDiffV": 0.00031,
  "ntpUtc": "2026-11-03T09:59:11Z",
  "raw": [4702, 32694, 32690, 1256],
  "rtcUtc": "2026-11-03T09:59:10Z",
  "sampleAgeMs": 3419,
  "schemaVersion": 4,
  "stm32": {
    "available": false, "echoIgnored": 0, "fresh": false,
    "messageCount": 0, "rejectedCount": 0,
    "reset": { "ackCount": 0, "ackKind": "none", "cooldownRemainingMs": 0,
               "remainingMs": 0, "state": "idle", "timeoutSeconds": 30 }
  },
  "timestampUtc": "2026-11-03T09:59:10Z",
  "volts": [0.36734, 2.55422, 2.55391, 0.09812]
}
```


What each part means, and what Terra does with it:

Field	Example	What it means, simply	What Terra does with it
timestampUtc	2026-11-03T09:59:10Z	When the reading was taken, in world time (UTC). In Dubai that is 13:59:10.	The reading's time. Everything is filed under it.
rtcUtc	2026-11-03T09:59:10Z	The same moment by the station's own clock.	Backup time if timestampUtc is missing.
ntpUtc	2026-11-03T09:59:11Z	Internet time when the record was put together (here 1 second later).	Second backup. A big gap from rtcUtc means the station clock is wrong ("Bad time" alarm).
device	aqms-monitor	Which station sent it.	Picks that station's set-up and calibration records.
volts	0.36734, 2.55422, 2.55391, 0.09812	The four sensor signals in volts: A0 = NH3 WE, A1 = H2S WE, A2 = H2S AE, A3 = VOC (PID).	The main input: × 1000 = mV, then the gas recipes.

Field	Example	What it means, simply	What Terra does with it
raw	4702, 32694, 32690, 1256	The same four signals as converter steps: 0 to 32,768 steps means 0 to 2,560 mV.	Backup if volts is missing: mV = steps × 2560 ÷ 32768. Also a cross-check.
a0Volts	0.36734	A0 (NH3) again, on its own.	Not needed: it is the same as the first value in volts.
h2sDiffV	0.00031	A1 minus A2, in volts (0.31 mV).	Not used: the H2S recipe takes each zero off first. A handy check only.
combinedString	2026-11-03,09:59:10,0.36734,2.55422,2.55391,0.09812,NA	The whole reading as one line of text: date, time, A0, A1, A2, A3, then the STM32 status (NA = not connected).	Keep it exactly as it arrived (the raw package text). Last backup for the time and the volts.
adcOk	true	The station's voltage converter (ADC) read the sensors correctly.	If false, the voltages cannot be trusted: no gas numbers, the reading is invalid.
averageCount	1	How many quick samples were averaged into this reading.	Diagnostics only.
sampleAgeMs	3419	How old the measurement was (3.4 seconds) when the record was made.	Diagnostics only. A very large value means old data.
batteryV	11.7587	The station's power supply voltage.	Goes to the "Power or box limit" technical alarm. Never used in a gas formula.

Field	Example	What it means, simply	What Terra does with it
schemaVersion	4	Which record layout this is.	Tells Terra how to read the record.
destination	client	Who the record is addressed to.	Not used.
stm32	available false, counts 0, reset state idle	The status of the link to the station's second small computer (the STM32 board), and of its reset command. Here it is not connected.	Diagnostics only. Not used in the gas maths.

The meanings of averageCount, destination and the stm32 block are taken from their names; the station firmware is the final word on them. None of them changes a gas number.

The four things Terra really needs from each record: the time (timestampUtc), the station (device), the four voltages (volts) and whether the converter worked (adcOk). Everything else is a backup or a health check of the equipment.

5. Temperature and humidity

Sensor temperature (measured at the top of each gas sensor): when a station sends it, Terra uses it for the temperature correction (see "Temperature correction"). Until the stations send it, Terra uses the **Open-Meteo hourly air temperature** at each station's own latitude and longitude instead, for the sensitivity part only. **Air temperature and humidity** from the station, if fitted, are stored but **not shown to DM** on the dashboard or in reports and are not used in the gas maths; they help with fault-finding, so show them only in an administrator's diagnostics screen.

Calibration: turning voltage into gas

A gas sensor does not give ppm or ppb. It gives a small voltage (mV). Calibration is the recipe that turns that voltage into a gas number.

We use Alphasense's method only. Every sensor needs two numbers: a **zero** (mV, what it reads when there is no gas) and a **sensitivity** (how many mV it gives per ppb or ppm). The **sensitivity** always comes from the sensor's own **factory calibration sheet**, made for that one sensor. The **zero** comes from a **site zero**: the sensor's own reading in clean air,

measured at the station, as Alphasense advise (see "Site zero" below). Until a sensor's factory sheet is available, Terra uses **Alphasense's published datasheet typical values** for that sensor type, clearly marked as indicative. Nothing else: no filtering, no smoothing, no extra corrections.

There are three recipes, one per gas.

Recipe 1: H2S (sensor H2S-B4, 4 electrodes)

The H2S sensor gives two voltages: WE (touches the gas) and AE (its twin that never touches the gas). The factory sheet gives a **WE zero**, an **AE zero** and the **sensitivity** in mV per ppb.

1. WE signal = WE − WE zero.
2. AE signal = AE − AE zero.
3. corrected = WE signal − AE signal.
4. **H2S (ppb) = corrected ÷ sensitivity**.

This is Alphasense's AAN 803-05 Algorithm 3 at 20 °C, where its temperature factor k′T is exactly 1. In Dubai heat that factor changes (see "Temperature correction" below), but that zero correction needs each sensor's own temperature and the board's electronic offsets, which are not available yet.

Worked example (made-up numbers, only to show the steps): WE = 300 mV, WE zero = 250 mV, AE = 280 mV, AE zero = 260 mV, sensitivity = 0.4 mV/ppb.

1. WE signal = 300 − 250 = 50 mV
2. AE signal = 280 − 260 = 20 mV
3. corrected = 50 − 20 = 30 mV
4. H2S = 30 ÷ 0.4 = **75 ppb**

Real example, from the test station's own H2S-B4 factory sheet: WE zero = 395 mV, AE zero = 362 mV, sensitivity = 1.727952 mV/ppb (1727.952 mV/ppm). So **H2S (ppm) = ((OP1 mV − 395) − (OP2 mV − 362)) ÷ 1727.952**, where OP1 is A1 and OP2 is A2. With OP1 = 0.600 V and OP2 = 0.300 V: (600 − 395) − (300 − 362) = 205 + 62 = 267 mV, and 267 ÷ 1727.952 = **0.1545 ppm** (154.5 ppb).

In use on the test station (site zero, see below): WE zero = 458 mV, AE zero = 359 mV, and the same sensitivity, 1.727952 mV/ppb. So **H2S (ppb) = ((A1 − 458) − (A2 − 359)) ÷ 1.727952**. Example: A1 = 500 mV and A2 = 360 mV give (42 − 1) ÷ 1.727952 = **23.7 ppb**.

Recipe 2: NH3 (sensor NH3-B1, 3 electrodes)

The NH3 sensor has only a WE (3 electrodes, no AE). Alphasense says: simply leave the AE out (this is AAN 803-05 Algorithm 4). The factory sheet gives a **WE zero** and the **sensitivity** in mV per ppm.

NH3 (ppm) = (WE − WE zero) ÷ sensitivity

Real example, from the test station's own factory sheet (sensor 77840031): WE zero = 328 mV, and sensitivity = 73.28 nA/ppm × 0.8 mV/nA = 58.624 mV/ppm. So **NH3 = (mV − 328) ÷ 58.624**. At 12:47 UTC on 2 November 2026 the station read 349.69 mV, so NH3 = 21.69 ÷ 58.624 = **0.37 ppm**.

In use on the test station (site zero, see below): WE zero = 420 mV, and the same sensitivity, 58.624 mV/ppm. So **NH3 (ppm) = (A0 − 420) ÷ 58.624**. Example: A0 = 449.31 mV gives 29.31 ÷ 58.624 = **0.50 ppm**. With the factory zero (328 mV), the station read about 1.4 ppm in a clean office, which was not real ammonia. With the site zero, clean air reads 0.

One honest warning for the screens: Alphasense's datasheet says this sensor's zero can be off by up to ±10 ppm in clean air. So a **rise** in NH3 is reliable, but a small NH3 number on its own is not. Label NH3 as **indicative**.

Recipe 3: VOC (sensor PIDX-A-040, the PID)

The factory sheet gives a **zero** (mV) and the **sensitivity** in mV per ppm. The PID reads a little lower when it is hot (see "Temperature correction").

VOC (ppm) = (V − V zero) ÷ sensitivity

Worked example (made-up numbers): V = 150 mV, V zero = 40 mV, sensitivity = 55 mV/ppm. Signal = 110 mV; VOC = 110 ÷ 55 = **2 ppm**.

In use on the test station (datasheet typical): V zero = 70 mV, sensitivity = 55 mV/ppm, so **VOC (ppm) = (A3 − 70) ÷ 55**. At 12:47 UTC on 2 November 2026 the station read 100.63 mV: 30.63 ÷ 55 = **0.557 ppm** at 20 °C. That hour Open-Meteo gave 32.1 °C at the Tridel office, so the PID sensitivity was 96.0 % and VOC = 30.63 ÷ (55 × 0.960) = **0.580 ppm**.

Two facts to show on screen, so nobody misreads VOC. A voltage **below** the zero (below 40 mV while the datasheet typical zero is in use) means the sensor is faulty, not that the air is extra clean — raise a sensor-fault alarm. And this sensor cannot see methane at all, so VOC is not "landfill gas". Label VOC as **indicative**.

Which calibration values we use

Sensor	Zero used	Sensitivity used	Where it comes from
H2S-B4 (test station)	WE 458 mV, AE 359 mV	1.727952 mV/ppb (1727.952 mV/ppm)	**Site zero** + factory-sheet sensitivity. The factory-sheet zeros (395 / 362 mV) stay as an older record.
NH3-B1+ (test station)	420 mV	58.624 mV/ppm	**Site zero** + sensitivity from factory sheet 77840031. The factory-sheet zero (328 mV) stays as an older record.
PIDX-A-040	70 mV (the middle of the datasheet's 40–100 mV range); below 40 mV = sensor fault	55 mV/ppm	**Datasheet typical**: Alphasense PIDX-A-040 datasheet

Datasheet typical values describe the sensor **type**, not this exact sensor, so the PID's own zero can differ by about ±0.55 ppm. That is why clean air can show a steady VOC reading (station history: about 0.63 ppm). Label VOC **indicative** until its factory sheet arrives; then add it as a **new** record with source "factory" and a later "valid from", and it takes over from that moment.

Site zero: what Alphasense told us

Alphasense measures each sensor's factory zero in filtered, scrubbed, dry air at 20–25 °C, on its own electronics. On our station the same sensors sit in real air, in more heat, on our own board, and their zero sits higher. With the factory zeros, a clean office read about **1.4 ppm NH3** and **38 ppb H2S**. Neither was real gas.

We asked Alphasense (November 2026). Their answer, in short:

- AAN 803-05 is the right and latest note. Correct the WE voltage first, then divide by the sensitivity. That is exactly what the recipes above do.
- NH3-B1 has no AE, so just leave the AE out of every formula.
- For outdoor air monitoring, take the zero from the **cleanest part of the day**, usually the early morning. Before deployment, zero the sensors in filtered, scrubbed, dry air.
- Then put the station next to a **reference station** (within about 1 m) for at least a week, and correct what is still wrong: an **offset** (it reads too high or too low all the time) and a **gain** (its peaks are too big or too small).

- No pressure correction is needed, as long as the sensors are calibrated at the pressure where they work. Dubai is at sea level.
- Alphasense has no humidity correction to give.

What we did on the test station. We took the middle (median) voltage between 02:00 and 05:00 Dubai time on five nights, 29 October to 3 November 2026 (899 readings): NH3 WE **420 mV**, H2S WE **458 mV**, H2S AE **359 mV**. These became new calibration records with source **"site zero"**. The sensitivities stayed as printed on the factory sheets, and the factory-zero records stay in the list for history.

What Terra should do. A site zero is simply a new calibration record (the rules below apply). Repeat it when a sensor or its board is changed, when a station moves, and before deployment (in zero air).

One thing to know. The test station reads highest at night and lowest in office hours. That is temperature, not gas. With a night-time zero, daytime values therefore show 0. Only the H2S zero correction, with each sensor's own temperature, can remove that swing; the sensitivity correction below does not.

Temperature correction: built in, needs the sensor temperature

Heat changes two things in a sensor: its **zero** and its **sensitivity**. The reference code and the Excel sheet correct both. Both need the temperature at the **top of each sensor** (to ±1 °C), which the station does not send yet. For the sensitivity, an outdoor air temperature is a usable stand-in (next paragraph); for the zero it is not, and the zero correction also needs the board's electronic offsets, so it stays off. At 20 °C the corrections change nothing.

For now, on the test website. Until the station measures its own sensor temperature, the website, the station history and the reference code's run_sample all use the **Open-Meteo hourly air temperature at the Tridel Dubai office** (QD01, DAFZA Industrial Park, Al Qusais Industrial Area 5). Open-Meteo answers from its nearest grid point, 25.318 N 55.394 E, which is 3.4 km from the office. It is used for the **sensitivity correction only**; the H2S zero correction stays off. The hourly temperatures used are in the package (Open_Meteo_hourly_temperature_Tridel_office.csv). This is outdoor air temperature: a sensor inside a box or an air-conditioned room can be several degrees different, so Terra should switch to each sensor's own temperature as soon as the stations send it. On the five landfill stations, use Open-Meteo at each station's own latitude and longitude. In the reference code, pass the value as `sensor_temp_c`.

1. Sensitivity, all three sensors. AAN 803-05 says to divide by the sensitivity at the sensor's temperature, and each datasheet has a chart of it. We read these values off the charts, so they are approximate; Alphasense may be able to send the exact numbers.

Sensor	0 °C	10 °C	20 °C	30 °C	40 °C	50 °C
H2S-B4 (datasheet Fig. 1)	92 %	96 %	100 %	103 %	105.5 %	105.5 %
NH3-B1 (datasheet Fig. 2)	100 %	98.5 %	100 %	103 %	110 %	125 %
PIDX-A-040 (ideal gas law)	107.3 %	103.5 %	100 %	96.7 %	93.6 %	90.7 %

So **sensitivity at T = sensitivity from the sheet × % ÷ 100**, and the gas is worked out with that sensitivity. The PID datasheet says its output follows the ideal gas law, so for the PID **% = 100 × 293.15 ÷ (273.15 + T)**: it reads about 6 % low at 40 °C. (An older Alphasense note, AAN 302-04, says PID temperature can be ignored, but that note is about a different PID model.)

2. Zero, H2S only. AAN 803-05 gives four algorithms and a factor table. In Algorithm 3 the factor k′T is +1 at 20 °C but **−1.2 at 30 °C and above**, so the AE gets added instead of taken away. In Algorithm 4 the factor k′′T is a **current in nA**: it matches the datasheet's zero chart (−65, −125 and −180 nA at 30, 40 and 50 °C), so it is multiplied by the board gain (mV per nA) before it is used. At 20 °C, Algorithm 3 gives exactly the same answer as Recipe 1. The zero correction also needs:

1. the board's two **electronic offsets** (WEe and AEe), from the ISB sheet ("Electronic Zero WE / AUX");
2. a chosen algorithm: AAN 803-05 Table 2 suggests Algorithm 1 for this sensor type, so compare with a reference station;
3. for Algorithm 4 only, the **board gain**, from the ISB sheet.

NH3 gets no zero correction: Table 3 has no row for NH3-B1, and its datasheet gives no zero data.

What a calibration record holds

Field	What it is	Used by
Station and channel	Which station and which gas (H2S, NH3 or VOC).	All
Sensor model and serial	As printed on the sensor and its factory sheet.	All
Factory sheet number and date	From the factory sheet.	All
Source	"factory", "site zero" or "datasheet". Show it on screen next to the value.	All
WE zero (or V zero for VOC), mV	From the site zero, or from the factory sheet.	All

Field	What it is	Used by
AE zero, mV	From the site zero, or from the factory sheet.	H2S only
Sensitivity	From the factory sheet: mV per ppb (H2S) or mV per ppm (NH3, VOC).	All
Electronic offsets WEe and AEe, mV	Optional. From the ISB sheet ("Electronic Zero WE / AUX"). Only needed for the H2S zero correction.	H2S
Temperature algorithm	Optional. AAN 803-05 algorithm 1, 2, 3 or 4. Empty = no zero correction. (The sensitivity correction needs only the sensor temperature.)	H2S
Board gain, mV per nA	Optional. From the ISB sheet. Only Algorithm 4 needs it, because its factor k''T is given in nA.	H2S
Valid from	The date and time this record starts to apply.	All
Entered by, entered at	Filled in by Terra from the login and the clock.	All
Notes	Free text. For a site zero: the dates, hours and number of readings used.	All

That is the whole record: at most three numbers per sensor (the zeros and the sensitivity), plus, for the H2S zero correction later, the optional electronic offsets, board gain and algorithm.

The eight calibration rules

1. A new calibration is always a **new** record. Never edit or overwrite an old one.
2. For each reading, use the record for that station and gas with the latest "valid from" that is **not after** the reading's time.
3. Never recalculate old data. A new calibration changes **future** readings only.
4. Records can never be deleted. A mistake is fixed by adding a corrected record with a note.
5. Every new record writes an audit entry: who, when, and the values typed.
6. Only the DM Administrator can add records, plus Tridel if Tridel keeps a login (see "Who can do what").
7. The list of all records can be exported as a table.

8. If a reading has **no** record to use, do not show a number. Mark the reading and raise the "calibration missing" alarm.

One exception, on the test station only: its site zero was applied back to its very first reading, so that the whole test history in the package is worked out the same way. On the five landfill stations, follow rule 3.

Terra does all the maths

The station sends only raw voltages, and **Terra does every calculation on the server**. For each reading, Terra:

1. finds the calibration record for that station and gas with the latest "valid from" that is not after the reading's time;
2. checks that no voltage is stuck at the end of its range (see "Technical alarms"); if one is, stores "sensor fault" and no number;
3. looks up the temperature for that reading (each sensor's own; until the stations send it, the Open-Meteo hourly air temperature at the station's location) and works out the sensitivity at that temperature;
4. applies the recipe above for that gas;
5. stores the raw voltages, the gas result, which calibration record was used, and the temperature and sensitivity used.

Keep the raw voltages for ever, next to the result. They are the proof of how each gas number was made, and they let a future factory sheet be checked against past data. The reference code in the package does exactly these steps, in Python and in JavaScript: one file each (aqms_reference.py and aqms_reference.js), with a plain-words comment on every step, so it can be copied straight into Terra.

Follow one record from start to finish

Here is the record from section 4a, step by step, exactly as Terra (and the reference code) handles it.

1. **Read the time and the station.** timestampUtc is 09:59:10 UTC, which is 13:59:10 in Dubai. device is aqms-monitor. adcOk is true, so the voltages can be used.
2. **Turn volts into millivolts** (× 1000): A0 = 367.34 mV, A1 = 2554.22 mV, A2 = 2553.91 mV, A3 = 98.12 mV. The converter steps agree: 32694 × 2560 ÷ 32768 = 2554.22 mV.
3. **Find the calibration record for each gas.** H2S: CAL-H2S-SITE-0001 (zeros 458 and 359 mV). NH3: CAL-NH3-SITE-0001 (zero 420 mV). VOC: CAL-VOC-DS-0001 (zero 70 mV).
4. **Check for a stuck channel.** A1 and A2 are both above 2550 mV, pressed against the top of the range (2560 mV). So H2S gets **"sensor fault" and no number**. Without this check the recipe would give ((2554.22 − 458) − (2553.91 − 359)) ÷ 1.727952 = −57 ppb,

which would show as **0 ppb, "Normal"**, and hide a broken sensor. A0 and A3 are fine.

5. **Look up the temperature.** The station sends none, so Terra takes Open-Meteo for that hour (09:00 UTC) at the Tridel office: **32.6 °C**. At 32.6 °C the NH3 sensor is 104.82 % as sensitive as at 20 °C, and the PID is 95.88 %.
6. **Work out each gas.**
   - NH3 = (367.34 − 420) ÷ (58.624 × 1.0482) = −0.86 ppm. Below 0 counts as 0, so **NH3 = 0 ppm**: the signal is below the site zero, which means clean air.
   - VOC = (98.12 − 70) ÷ (55 × 0.9588) = **0.533 ppm** (it would be 0.511 ppm without the temperature correction).
   - H2S: no number (step 4).
7. **Store everything.** The record exactly as it arrived, the four millivolt values, the three results (or the reason there is none), which calibration record was used, the temperature (32.6 °C) and the sensitivity used.
8. **Averages and alarms.** The reading goes into the 09:50–10:00 UTC window. Its NH3 and VOC count towards the averages as normal. Its H2S does not count; on 3 November H2S was stuck from 06:47 to 12:41 UTC, so every H2S window in that time is invalid. The stuck channel raises the **"Channel stuck" technical alarm** for DM. The battery voltage (11.76 V) goes to the power check.

Averages, and good or bad data

A single 2-minute reading jumps around too much to trust. So Terra makes a **10-minute average** from the five readings in each 10 minutes, and alarms only ever look at that average.

When is a reading good?

A reading is **valid** (good) only if all five of these are true:

1. Its time makes sense (not in the future, not repeated).
2. The sensor status says normal.
3. The value is inside the allowed range.
4. The station is **not** warming up, being calibrated, or under maintenance.
5. No voltage it needs is stuck at the end of its range (see "Technical alarms").

If any one is false, the reading is **invalid**. Keep it, with a reason code saying why, but leave it out of every average and every alarm.

The 10-minute average

- There should be **5** readings in 10 minutes.

- At least **4** must be valid. Then the average = add the valid ones and divide by how many there are.
- If fewer than 4 are valid, the 10-minute value is marked **invalid** and cannot start an alarm.

Example: the five readings are 10, 12, (bad), 11, 13. Four are valid, so the average is (10 + 12 + 11 + 13) ÷ 4 = **11.5**. If two had been bad, there would be only 3 valid readings, so that 10 minutes is invalid.

Longer averages

Terra also makes 1-hour, 8-hour and 24-hour averages from the same valid data. The 24-hour average is **rolling**: at any moment it covers the 24 hours just gone. For a report period, if less than **75 %** of the expected readings are valid, still show the result but clearly mark it **incomplete**.

Five rules that are never broken

- **Missing stays missing.** Never fill a gap with zero. Never copy the last value forward. A chart shows a gap.
- **Bad data is kept, not deleted.** It carries its reason code forever.
- **The quality flag goes everywhere** the number goes: every screen, every export.
- **Late data keeps its real time.** A package that arrives late keeps both its measurement time and its arrival time, so it lands in the right 10 minutes.
- **A negative result counts as 0.** If a formula gives a value below 0, Terra shows and stores **0**, and uses 0 in every average (10-minute, 1-hour, 8-hour, 24-hour). This stops bad-looking values on screen. The raw voltage is still kept as it arrived.

Wind is averaged differently

Wind **speed** is a normal average. Wind **direction** cannot be. The average of 350° and 10° must be 0° (north), but a normal average gives 180° (south) — exactly backwards. So direction uses a **vector average**:

1. For each valid direction, work out its sine and its cosine.
2. Average all the sines. Average all the cosines.
3. Average direction = atan2(average sine, average cosine), turned into degrees. If the answer is below 0, add 360.

Check your code with this: directions 350° and 10° must give **0°**, and 80° and 100° must give **90°**.

Alarms: when the system shouts

An alarm means "this gas is higher than normal **at this station**". There are three levels, from mild to serious: **Warning**, **Action** and **Critical**. Alarms only ever look at **valid 10-minute averages**, never at raw 2-minute readings.

Step 1: learn what "normal" is (the baseline)

Nobody knows the normal smell level at each station yet. So after the stations are switched on, they run for **72 hours**, then a **30-day** baseline review. From the valid 10-minute averages of that period, for each station and each gas, work out two numbers:

- **B** = the average (the normal level).
- **S** = the standard deviation (how much it normally wobbles up and down).

Until the 30 days are over there are no gas thresholds, so no gas alarms. Technical alarms (further below) work from day one.

Step 2: set the three thresholds

Tridel (or the DM Administrator, see "Who can do what") also sets an **alarm floor** for each station and gas: a lowest-allowed threshold, so sensor noise near zero can never cause an alarm. Then:

Level	Threshold = the bigger of these two
Warning	alarm floor, or B + 2 × S
Action	alarm floor, or B + 3 × S
Critical	alarm floor, or B + 4 × S

Example: (made-up numbers) at AQMS-02, H2S has B = 5 ppb, S = 2 ppb, and the floor is 8 ppb. Warning = bigger of 8 and 9 = **9 ppb**. Action = bigger of 8 and 11 = **11 ppb**. Critical = bigger of 8 and 13 = **13 ppb**.

Thresholds are set **per station and per gas**, not one for all. Terra stores the numbers the Administrator types in; it does not change them by itself. Each threshold has a "valid from" date, so an old alarm always keeps the threshold that applied at its time.

Step 3: when does an alarm start?

Level	It starts when	What Terra does
Warning	2 valid 10-minute averages **in a row** are above Warning, **or** 1 valid 30-minute average is above it.	Create a Warning event and notify straight away.
Action	2 valid 10-minute averages in a row are above Action, **or** 1 valid 30-minute average is above it, **or** a rapid rise (below).	Create an Action event and notify straight away.
Critical	Just **1** valid 10-minute average above Critical. Do not wait for a second one.	Create a Critical event and notify straight away.

An **invalid** 10-minute window is simply skipped. It does not count towards "in a row", and it does not break the row either.

Rapid rise starts an Action alarm when both are true: the newest valid 10-minute average is at least **twice** the middle value (median) of the three valid averages before it, **and** it is above Warning.

Example: the last three averages were 4, 6 and 5, so the median is 5. The new average is 12. 12 is at least 2 × 5 = 10, and 12 is above the Warning of 9. So: **Action**.

One alarm at a time

Each station and gas has **only one** alarm open at a time, held at the highest level reached. If a Warning is open and Action is reached, the same alarm is **raised to** Action. It is not a second alarm.

Step 4: when does an alarm stop?

- **Warning** clears after **3 valid 10-minute averages in a row** are below the Warning threshold.
- **Action and Critical** need the same 3 in a row below, **and** a person must have acknowledged them.
- Clicking "acknowledge" alone never clears an alarm.
- Every start, level change, acknowledgement and clear is written to the alarm history, with the user and the time.

Step 5: who is told, and how fast they must look

Level	Notification	Acknowledge within	What the DM officer checks
Warning	In the app, at once	4 hours	The trend, data quality, wind, and the nearby stations
Action	In the app at once, and escalated	2 hours	Start an investigation and confirm what is happening
Critical	Urgent escalation at once	30 minutes	Urgent review under the DM escalation procedure

The main person told is the **DM Operator**, inside the app. For email:

- Send one email when an alarm starts, one when it goes up a level, and one when it clears. Do not repeat while it stays on.
- Wait at least **60 minutes** between emails for the same station and gas, so a flickering alarm cannot flood mailboxes.
- The email says: station, gas, level, value, threshold, averaging time and time. It does **not** explain or guess the cause.
- Email only works if DM IT allows outgoing mail and gives a mail relay. If not, alarms stay in the app and the log only.

Technical alarms

These are about the **equipment**, not the air. They show in Terra for **DM**, clearly marked as equipment alarms. They are never shown as a gas alarm and never called an "exceedance". When one appears, DM tells Tridel so the equipment can be checked or fixed. **No data is sent to Tridel.**

Technical alarm	Starts when	Clears when
Communication lost	No 10-minute package for **30 minutes** after the last good one.	A package arrives. The station then sends the missed readings.
Sensor fault or out of range	The sensor reports a fault, or a value is outside its allowed range.	Readings are normal again.

Technical alarm	Starts when	Clears when
Flat line	A gas shows the exact same value for more than **60 minutes**, with no reason.	The value changes.
Bad time	A timestamp is old, repeated, or the station clock is wrong.	Times are correct again.
Power or box limit	Power or the enclosure goes past its limit.	Back within limits.
Low data completeness	A 10-minute or longer window has too few valid readings.	The next window has enough.
Calibration missing	A reading arrives with no calibration record to use.	The Administrator adds a record.
Channel stuck	A voltage is 2550 mV or more (the top of the range is 2560 mV), or 5 mV or less: the sensor is unplugged, has no power, or is shorted. Show "Sensor fault" and no number. For H2S, A1 or A2 alone is enough.	The voltage is back inside its range.

During the baseline period, odd behaviour such as drift, sudden jumps, noise or flat lines is shown to DM as an **equipment alarm**, not a gas alarm. DM can then ask Tridel for a spare-sensor drift check.

Why "Channel stuck" needs its own rule. On 3 November 2026 the test station's A1 and A2 both sat at 2560 mV for six hours (06:47 to 12:41 UTC). In the H2S formula they cancel each other out, so without this rule H2S showed **0 ppb and "Normal"** while the sensor was not working at all.

Events: what happens next

There are two kinds of event. An **alarm event** is made by Terra by itself when an alarm starts. A **complaint event** is made by a DM officer when someone reports a bad smell.

Kind 1: alarm events

Every alarm event stores the station, gas, value, threshold, averaging time, quality status and time. It then moves through these states:

State	What it means	How it gets there
Raised	The start rule was met.	By itself
Acknowledged	A person has seen it. The alarm is **still on**.	A person clicks; Terra saves who and when
Escalated	A higher level was reached while it was on. The same alarm moves up; no new alarm is made.	By itself
Cleared	The stop rule was met.	By itself, with the clear time
Closed	It is both cleared **and** acknowledged.	By itself, once both are true

Operators can add notes to an alarm event. Nothing in its history is ever deleted.

Kind 2: complaint events (the investigation)

This is the most important screen for DM. A person phones DM about a bad smell. A DM officer records it, looks at what the stations measured around that time, and writes down what they decided. It is two forms, one review screen and one report.

Step 1: Register the complaint. The officer fills in a form:

Field	What to enter
DM reference	DM's own complaint number (required).
Complaint time	When the person says they smelled it.
Location	The area or address they gave.
What they smelled	Type of smell, how strong, how long.
Window start	Defaults to **6 hours before** the complaint time. Can be changed.
Window end	Defaults to **2 hours after** the complaint time. Can be changed.
Stations	All five, by default.
Data to show	Raw 2-minute, 10-minute or 1-hour averages.
Logged by, at	Filled in by Terra.

On save, the event is **Open** and the review screen opens.

Step 2: Review screen. One screen, all on the same time line across the saved window:

- **Gas panel:** H2S, NH3 and VOC for every chosen station.
- **Wind panel:** wind speed and direction.
- **Quality strip:** for each station, when it was online or offline, completeness, and any invalid windows.
- **Summary table:** for each station and gas, the minimum, average and maximum, and how many valid windows there were.
- **Complaint line:** a vertical line at the complaint time on every panel.

Opening the event again must show exactly the same window and stations. Terra does **no** maths here except averaging: no ranking of stations, no upwind or downwind guess, no blaming a source. The officer decides, not Terra.

Step 3: Record the outcome. This can be done hours or days later. The officer fills in:

Field	Required?	What it is
Source classification	Yes	Pick one: Probable, Possible, Unlikely or Unconfirmed.
Justification	Yes	Why they picked that.
Conclusion	Yes	What they concluded.
Follow-up needed	Yes	Yes or no, with a due date if yes.
Operational information reviewed, and its source	No	What DM site information they looked at, and where it came from.
Limitations	No	Anything that made the check harder.
Action taken	No	What was done.

On save, the event moves to **Under review**.

Step 4: The report. Terra makes a PDF from a template: the event details, the saved window, pictures of the review panels, the summary table, a statement of how complete the data was, and every field the officer filled in. The data behind it can be downloaded as CSV. Once the report is made, the event is **Closed**.

Complaint state	Meaning
Open	Registered and window saved; no outcome yet.
Under review	The officer has started writing the outcome.

Complaint state	Meaning
Closed	Outcome written and report made.

Who can do what

The line is simple. A **DM Operator** can see everything and investigate, but cannot change anything that changes what a number means. Thresholds, calibration and station set-up belong to the **DM Administrator**.

Can they…	DM Operator	DM Administrator
See the dashboard, time series, wind rose, pollution rose, heat map, histogram, box plot	Yes	Yes
Make reports and export data	Yes	Yes
Acknowledge alarms	Yes	Yes
Register a complaint, review it, record the outcome, export its report	Yes	Yes
Change alarm thresholds and levels	No	Yes
Add or correct calibration values	No	Yes
Set up stations, averages, data keeping time and alarm recipients	No	Yes
See the raw package log and diagnostics (including temperature and humidity)	No	Yes
Manage users	No	Yes

All data and all alarms, including technical alarms, stay with DM in Terra. When equipment needs fixing, DM tells Tridel. No data is sent to Tridel.

Tridel's login is not decided yet, so build Terra to allow both options:

	Option A: Tridel keeps a login	Option B: no Tridel login
Tridel account	A separate administrator account during the support period. Not shown to DM, not in the DM user manual.	None.

	Option A: Tridel keeps a login	Option B: no Tridel login
Who types in calibration numbers	Tridel or the DM Administrator	DM Administrator only; Tridel sends the numbers to DM
Who sets the alarm floor	Tridel, inside Terra	DM Administrator, using the value Tridel recommends

The simple way to support both: make "Tridel administrator" an ordinary user account that DM can switch on or off. Everything else stays the same. To be confirmed with Akhil Narayanan.

Where the two DM documents disagree

We have two documents for this project, and on a few points they say different things. Where they disagree, follow the **Monitoring Criteria** document, because it is the one written for Dubai Municipality's approval. Where only one of them talks about something, follow that one.

- **Monitoring Criteria** = "TriSense AQMS – Monitoring Criteria, Averaging Methodology and TEMS Alarm Configuration".
- **Functional Specification** = "AQMS Functional Specification".

Topic	Monitoring Criteria says	Functional Specification says	Build this
Names of the levels	Warning, Action, Critical	Advisory, Warning, High	Warning, Action, Critical
How a threshold is worked out	Bigger of the alarm floor and B + 2S, 3S or 4S	Administrator reads it off the histogram and types it in	Administrator types it in, using the Monitoring Criteria formula; the histogram screen shows B and S to help
How many windows before an alarm starts	2 in a row (or one 30-minute average); Critical needs only 1	3 in a row by default	Monitoring Criteria

Topic	Monitoring Criteria says	Functional Specification says	Build this
How an alarm stops	3 valid windows in a row below the threshold; Action and Critical also need an acknowledgement	3 windows below (threshold minus 10 %)	Monitoring Criteria (no 10 % gap)
Communication lost after	30 minutes	10 minutes	30 minutes
How often a station sends	Every 10 minutes, carrying five 2-minute readings	A packet every 2 minutes	Every 10 minutes
How fast someone must acknowledge	4 h, 2 h, 30 min	Not given	Monitoring Criteria
Email rules	Not given	Once on start, level-up and clear; 60 minutes apart	Functional Specification
Complaint investigation	Not given	Full workflow	Functional Specification
Calibration values to use	Not given	Zero, sensitivity and temperature coefficients from the manufacturer	**Alphasense method only**: the sensitivity from the factory sheet, the zero from a site zero, and Alphasense's own temperature factors (AAN 803-05 and the datasheets). Until a sheet arrives, Alphasense datasheet typical values. (Decided by Tridel.)

Topic	Monitoring Criteria says	Functional Specification says	Build this
Where technical alarms go	Tridel support	Not given	DM, inside Terra. DM tells Tridel when equipment needs fixing. No data goes to Tridel. (Decided by Tridel.)

I have not yet confirmed these choices with Akhil Narayanan. If he decides differently on any row, that row changes; nothing else does.

Every number we use, and why

Every number below has a reason, and the reason is written next to it. Nothing was tuned to make the results look nicer.

Reading the station

Number	Where it is used	Why this number
× 1000	volts to millivolts	1 volt is 1,000 millivolts. All recipes work in mV.
2560 and 32,768	converter steps to mV	The station's converter measures 0 to 2.56 V in 32,768 equal steps, so one step is 0.078 mV.
2550 mV and 5 mV	"Channel stuck"	In clean air the sensors give roughly 90 to 500 mV, and gas pushes them up smoothly (the highest seen on the test station was about 1,430 mV). A voltage pressed against the very top (2560 mV) or bottom (0 mV) means a loose cable, no power or a short, not gas. 2550 leaves a 10 mV margin below the top.

Turning millivolts into gas

Number	Where it is used	Why this number
458 and 359 mV	H2S zeros (WE and AE)	The station's own H2S readings in clean air: the middle value between 02:00 and 05:00 Dubai time over five nights, as Alphasense advised.

Number	Where it is used	Why this number
420 mV	NH3 zero	The same, for NH3.
395 / 362 mV and 328 mV	old factory zeros, kept for history	Measured by Alphasense in clean, dry air at 20 °C on its own board. On our station they gave about 38 ppb H2S and 1.4 ppm NH3 in clean air, which was not real gas.
1.727952 mV per ppb	H2S sensitivity	Printed on the H2S sensor's own factory sheet.
58.624 mV per ppm	NH3 sensitivity	From factory sheet 77840031: 73.28 nA per ppm × 0.8 mV per nA (the board's gain).
70 mV and 55 mV per ppm	VOC zero and sensitivity	From the PID datasheet: its zero is somewhere between 40 and 100 mV (we take the middle, 70), and it typically gives 55 mV per ppm. Used until the PID's own sheet arrives.
40 mV	VOC "sensor fault"	The PID datasheet says a working PID never goes below 40 mV.
0	negative results	There can never be less than no gas. A small negative number is only noise around zero, so it counts as 0, also in every average.
±10 ppm and ±0.55 ppm	"indicative" labels for NH3 and VOC	From the datasheets: how far the NH3 zero and the PID zero can be off. That is why small NH3 and VOC numbers are shown as indicative.

Temperature

Number	Where it is used	Why this number
20 °C	the reference temperature	All factory numbers are measured at 20 °C, so at 20 °C nothing is corrected.

Number	Where it is used	Why this number
the % tables (for example 103 % at 30 °C)	sensitivity at the sensor's temperature	Read off each datasheet's chart of sensitivity against temperature (AAN 803-05 tells us to use them).
293.15	the PID's correction	20 °C on the absolute (kelvin) scale. Warm air is thinner, so fewer gas molecules reach the PID: its datasheet says it follows the ideal gas law.
−1.2, −180 and the other Table 3 factors	the H2S zero correction (off for now)	Copied from AAN 803-05 Table 3. The k''T factors are currents in nA, so they are multiplied by the board's gain.
32.6 °C (example) and 25.318 N, 55.394 E	the stand-in temperature	Open-Meteo's hourly air temperature at its weather point nearest the Tridel office (3.4 km away), until the station measures its own sensor temperature.

Averages and data quality

Number	Where it is used	Why this number
2 minutes, 5 readings, at least 4 valid	the 10-minute average	DM's Monitoring Criteria. 4 of 5 (80 %) keeps the average honest without throwing a window away for one bad reading. The test station sends one reading a minute, so there it is 8 of 10.
75 %	report completeness	DM's Monitoring Criteria. Below it, a report period is still shown but marked incomplete.
30 minutes	"Communication lost"	DM's Monitoring Criteria. Long enough to ride out a short network drop.
60 minutes	"Flat line"	Real air never stays at exactly the same value for an hour, so this means a stuck sensor or logger.

Alarms and events

Number	Where it is used	Why this number
72 hours, then 30 days	the baseline	DM's plan: enough data to learn what is normal at each station before any gas alarm.
B + 2S, 3S, 4S	Warning, Action, Critical	DM's Monitoring Criteria: 2, 3 or 4 "normal wobbles" (S) above the normal level (B).
alarm floor	the lowest allowed threshold	So tiny noise near zero can never start an alarm. Set per station and gas by Tridel or the DM Administrator.
2 in a row (Critical: 1)	alarm start	One odd window should not start an alarm, but a Critical level cannot wait for a second one.
2 × the median of the last 3	rapid rise	A sudden doubling above the recent normal is worth an Action alarm even before two windows pass.
3 in a row below	alarm clear	Stops an alarm switching on and off over and over ("flickering").
4 h, 2 h, 30 min	time to acknowledge	DM's Monitoring Criteria for Warning, Action and Critical.
60 minutes	gap between emails	Functional Specification: so a flickering alarm cannot flood mailboxes.
6 h before, 2 h after	complaint review window	Functional Specification defaults: the smell may have started well before the call.
UTC + 4	screen times	Dubai time. Everything is stored in UTC and shown in Dubai time.

Still open, and the files shared

None of these stop the Terra team building. Every one of them only fills in a number or a setting later.

- [ ] **Factory sheet for VOC.** The H2S-B4 and NH3-B1+ factory-sheet sensitivities are in use, with site zeros. Only the PIDX-A-040 still uses Alphasense datasheet typical values (indicative). Find its factory calibration sheet and add it as a new record.
- [x] ~~**Who does the gas maths:** decided. Terra does all of it, on the server.~~
- [ ] **Baseline.** 72 hours, then 30 days after the stations are switched on. Only then can B, S, the alarm floor and the three thresholds be typed in.
- [ ] **Email.** DM IT must allow outgoing mail and give a mail relay. Until then, alarms are in the app only.
- [ ] **Data keeping time** for raw and averaged data, to be agreed with DM IT.
- [ ] **The disagreements table above**, to be confirmed with Akhil Narayanan.
- [ ] **Tridel login in Terra:** Option A (Tridel keeps an administrator login) or Option B (no Tridel login). Build for both; to be confirmed with Akhil Narayanan.
- [ ] **Sensor temperature and electronic offsets.** To switch on temperature correction, each station must send the temperature at the top of each sensor, and, for the H2S zero, the ISB sheets' electronic offsets (WEe, AEe) and board gain must be entered. Until then, the Open-Meteo stand-in corrects only the sensitivity, and NH3 and H2S still swing with the day's temperature.
- [ ] **Zero air and a reference station.** Before deployment, zero each station in filtered, scrubbed, dry air (a new site-zero record). Then, if DM can arrange it, a week next to a DM reference station, to correct the offset and gain of each gas.
- [ ] **Test station H2S cable.** A1 and A2 were stuck at 2560 mV on 3 November from 06:47 to 12:41 UTC. Check the H2S sensor's cable and power before the stations are built.
- [ ] **Exact sensitivity-vs-temperature numbers.** The values in the code and Excel are read off the datasheet charts. Ask Alphasense for the numbers behind those charts (H2S-B4 Fig. 1, NH3-B1 Fig. 2, PIDX-A-040 Fig. 2) and replace them.

Files shared with this note

Everything is in one package: `AQMS_Terra_Handover_Package.zip`.

Folder	File	What it is for
01 Handover Note	AQMS_Handover_Note.pdf	This note, as a PDF.

Folder	File	What it is for
02 Dubai Municipality requirements	AQMS_Monitoring_Criteria_TEMS_Alarm_Configuration.pdf	The alarm levels, averaging and data-validity rules. Wins wherever the two DM documents disagree.
02 Dubai Municipality requirements	AQMS_Functional_Specification.docx	The screens, user roles, calibration record and complaint investigation steps.
03 Alphasense	Alphasense_AAN-803-05.pdf	Alphasense's note on the electrochemical sensors (H2S and NH3): the formulas, the site zero ("Initial calibration") and the temperature algorithms (Tables 1–3).
03 Alphasense	Alphasense_AAN-302-04.pdf	Alphasense's note for an older PID model: the zero rule and the error levels. Its temperature advice does not apply to the PIDX-A-040.
03 Alphasense	H2S-B4, NH3-B1 and PIDX-A-040 datasheets	Ranges, typical sensitivity and other sensor facts.

Folder	File	What it is for
04 Templates and test values	AQMS_Formulas_and_Templates.xlsx	Every formula in this note as a live Excel sheet with the worked examples, including the site zero and the temperature calculator, plus empty templates for the calibration record, threshold record and complaint event. Your code must give the same answers as the green cells.
04 Templates and test values	Live_station_data_2026-11-02.csv	300 live readings from the test station, 2 November 2026 (current packet style, one reading a minute; raw voltages only).
04 Templates and test values	Sample_station_data_test_unit.csv	560 older readings from 28 October 2026. Gas columns recalculated the same way as the history (site zeros, Open-Meteo temperature).

Folder	File	What it is for
04 Templates and test values	Station_history_readings.csv	The station's complete history, 28 October to 3 November 2026 (7,350 readings), recalculated: H2S and NH3 with their site zeros and factory-sheet sensitivities, VOC with datasheet typical values (indicative), and each sensitivity corrected to the Open-Meteo hourly temperature at the Tridel Dubai office. 198 H2S readings on 3 November are sensor faults.
04 Templates and test values	Station_history_10min_averages.csv	The same history as 10-minute averages (756 windows), at least 8 of 10 readings valid per window.
04 Templates and test values	Open_Meteo_hourly_temperature_Tridel_office.csv	The hourly air temperature used for the history: Open-Meteo grid point 25.318 N, 55.394 E, the nearest to the Tridel Dubai office.

Folder	File	What it is for
04 Templates and test values	Station_record_example_schema4.json	One real record exactly as the test station sends it today (3 November 2026, 09:59:10 UTC), explained in section 4a and worked through in "Follow one record".
05 Reference Code	Python and JavaScript	One file per language, aqms_reference.py and aqms_reference.js, ready to copy and paste, with a plain-words comment on every step. It does exactly what this note says, using only Alphasense's formulas, with the same 47 checks in each language; both give identical results on the live data, and run_sample uses the Open-Meteo temperature. It replaces the earlier airwatch-dubai engine; none of its filtering or smoothing is kept. Start with its README.

If something here is still unclear, please add a comment on this doc rather than waiting — I will answer it when I am back, and anything marked open above can be filled in without changing what you have built.